ANALYSIS REPORT

Wave 2.3 Residual Harmonic Temporal Hybrid Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the approved `Wave 2.3` residual harmonic temporal hybrid campaign. The campaign tested residual harmonic `GRU` and residual harmonic `LSTM` sequence regressors across `global`, `Fw`, and `Bw` direction surfaces, with three harmonic-basis tiers per surface: sparse `RCIM`, dense `0..240`, and dense `0..360`.

The campaign completed all 18 planned runs with zero launcher failures. The campaign winner by the configured selection policy, minimum `test_mae` with `test_rmse`, `val_mae`, and trainable parameter count as tie breakers, is `te_residual_harmonic_gru_sequence_remote_Fw_sparse_rcim`.

Campaign Artifacts

Artifact	Path
Campaign output	`output/training_campaigns/2026-05-27-18-55-47_wave2c_residual_harmonic_temporal_hybrid_campaign_2026_05_27`
Campaign leaderboard	`output/training_campaigns/2026-05-27-18-55-47_wave2c_residual_harmonic_temporal_hybrid_campaign_2026_05_27/campaign_leaderboard.yaml`
Best-run pointer	`output/training_campaigns/2026-05-27-18-55-47_wave2c_residual_harmonic_temporal_hybrid_campaign_2026_05_27/campaign_best_run.yaml`
Execution report	`output/training_campaigns/2026-05-27-18-55-47_wave2c_residual_harmonic_temporal_hybrid_campaign_2026_05_27/campaign_execution_report.md`
Planning report	`doc/reports/campaign_plans/wave_2/2026-05-27-18-08-32_wave2c_residual_harmonic_temporal_hybrid_campaign_plan_report.md`
Model technical document	`doc/technical/2026-05/2026-05-27/2026-05-27-18-08-32_wave2c_residual_harmonic_temporal_hybrids.md`
Remote launcher standard	`doc/technical/2026-05/2026-05-27/2026-05-27-18-35-06_campaign_launcher_remote_execution_standard.md`
Closeout technical document	`doc/technical/2026-05/2026-05-28/2026-05-28-11-35-34_wave2c_campaign_closeout.md`

Execution Summary

Field	Value
Campaign name	`wave2c_residual_harmonic_temporal_hybrid_campaign_2026_05_27`
Started at	`2026-05-27T18:55:47`
Finished at	`2026-05-27T22:35:20`
Completed runs	18
Failed runs	0
Tested model types	`residual_harmonic_gru_sequence` , `residual_harmonic_lstm_sequence`
Tested direction scopes	`global` , `Fw` , `Bw`
Harmonic basis tiers	`sparse_rcim` , `dense_240` , `dense_360`
Sparse harmonic list	`[0, 1, 3, 39, 40, 78, 81, 156, 162, 240]`

Campaign Winner

Field	Value
Run name	`te_residual_harmonic_gru_sequence_remote_Fw_sparse_rcim`
Run instance	`2026-05-27-19-07-31__te_residual_harmonic_gru_sequence_remote_fw_sparse_rcim`
Model family	`residual_harmonic_gru_sequence_fw_sparse_rcim`
Model type	`residual_harmonic_gru_sequence`
Direction scope	forward only
Trainable parameters	151,060
Validation MAE	0.003309
Validation RMSE	0.003828
Test MAE	0.003200
Test RMSE	0.003635

The winning checkpoint is stored at `output/training_runs/residual_harmonic_gru_sequence_fw_sparse_rcim/2026-05-27-19-07-31__te_residual_harmonic_gru_sequence_remote_fw_sparse_rcim/checkpoints/residual_harmonic_gru_sequence-epoch=015-val_mae=0.00330934.ckpt` .

Leaderboard

Rank	Run	Family	Scope	Test MAE	Test RMSE	Val MAE	Params
1	`te_residual_harmonic_gru_sequence_remote_Fw_sparse_rcim`	`residual_harmonic_gru_sequence_fw_sparse_rcim`	Fw	0.003200	0.003635	0.003309	151,060
2	`te_residual_harmonic_gru_sequence_remote_Fw_dense240`	`residual_harmonic_gru_sequence_fw_dense240`	Fw	0.003219	0.003653	0.003270	151,522
3	`te_residual_harmonic_lstm_sequence_remote_Fw_sparse_rcim`	`residual_harmonic_lstm_sequence_fw_sparse_rcim`	Fw	0.003234	0.003679	0.003344	201,364
4	`te_residual_harmonic_gru_sequence_remote_Fw_dense360`	`residual_harmonic_gru_sequence_fw_dense360`	Fw	0.003241	0.003677	0.003265	151,762
5	`te_residual_harmonic_lstm_sequence_remote_Fw_dense240`	`residual_harmonic_lstm_sequence_fw_dense240`	Fw	0.003262	0.003706	0.003307	201,826
6	`te_residual_harmonic_lstm_sequence_remote_Fw_dense360`	`residual_harmonic_lstm_sequence_fw_dense360`	Fw	0.003351	0.003774	0.003302	202,066
7	`te_residual_harmonic_lstm_sequence_remote_global_sparse_rcim`	`residual_harmonic_lstm_sequence_sparse_rcim`	global	0.003368	0.003808	0.003632	201,364
8	`te_residual_harmonic_lstm_sequence_remote_Bw_sparse_rcim`	`residual_harmonic_lstm_sequence_bw_sparse_rcim`	Bw	0.003440	0.004030	0.003764	201,364
9	`te_residual_harmonic_gru_sequence_remote_global_sparse_rcim`	`residual_harmonic_gru_sequence_sparse_rcim`	global	0.003440	0.003848	0.003607	151,060
10	`te_residual_harmonic_gru_sequence_remote_Bw_dense360`	`residual_harmonic_gru_sequence_bw_dense360`	Bw	0.003468	0.004050	0.003773	151,762
11	`te_residual_harmonic_lstm_sequence_remote_global_dense240`	`residual_harmonic_lstm_sequence_dense240`	global	0.003473	0.003925	0.003624	201,826
12	`te_residual_harmonic_lstm_sequence_remote_global_dense360`	`residual_harmonic_lstm_sequence_dense360`	global	0.003477	0.003940	0.003648	202,066
13	`te_residual_harmonic_gru_sequence_remote_Bw_dense240`	`residual_harmonic_gru_sequence_bw_dense240`	Bw	0.003492	0.004074	0.003585	151,522
14	`te_residual_harmonic_gru_sequence_remote_Bw_sparse_rcim`	`residual_harmonic_gru_sequence_bw_sparse_rcim`	Bw	0.003502	0.004061	0.003833	151,060
15	`te_residual_harmonic_gru_sequence_remote_global_dense240`	`residual_harmonic_gru_sequence_dense240`	global	0.003511	0.003983	0.003600	151,522
16	`te_residual_harmonic_gru_sequence_remote_global_dense360`	`residual_harmonic_gru_sequence_dense360`	global	0.003535	0.003999	0.003628	151,762
17	`te_residual_harmonic_lstm_sequence_remote_Bw_dense360`	`residual_harmonic_lstm_sequence_bw_dense360`	Bw	0.003556	0.004125	0.003729	202,066
18	`te_residual_harmonic_lstm_sequence_remote_Bw_dense240`	`residual_harmonic_lstm_sequence_bw_dense240`	Bw	0.003605	0.004129	0.003742	201,826

Technical Interpretation

The strongest Wave 2.3 scalar training result is the forward-only sparse `RCIM` residual harmonic `GRU`, with test MAE 0.003200. The dense harmonic branches did not improve the campaign winner: the second ranked model is the forward-only dense `0..240` residual harmonic `GRU`, and the third ranked model is the forward-only sparse `RCIM` residual harmonic `LSTM`.

This result is execution-valid and useful as a model-family probe, but it does not exceed the current Wave 2.2 scalar training winner. The program-level scalar best remains `te_periodic_gru_sequence_remote_Bw` with test MAE 0.002344. That means Wave 2.3 should be retained as a completed comparison branch, while Wave 2.2 remains the stronger scalar training baseline until a future campaign or official `TE Curve Verification Pipeline` review changes that conclusion.

The campaign also shows that adding a recurrent residual branch over an explicit harmonic base is not automatically superior to feeding the harmonic prior directly into the recurrent temporal model. That is a useful negative result: the residual decomposition remains inspectable, but the sparse periodic recurrent Wave 2.2 formulation is still the better scalar candidate.

Registry Effects

Registry Scope	Current Family Best	Test MAE	Interpretation
`residual_harmonic_gru_sequence_sparse_rcim`	`te_residual_harmonic_gru_sequence_remote_global_sparse_rcim`	0.003440	Updated by this campaign
`residual_harmonic_gru_sequence_fw_sparse_rcim`	`te_residual_harmonic_gru_sequence_remote_Fw_sparse_rcim`	0.003200	Updated by this campaign
`residual_harmonic_gru_sequence_bw_sparse_rcim`	`te_residual_harmonic_gru_sequence_remote_Bw_sparse_rcim`	0.003502	Updated by this campaign
`residual_harmonic_gru_sequence_dense240`	`te_residual_harmonic_gru_sequence_remote_global_dense240`	0.003511	Updated by this campaign
`residual_harmonic_gru_sequence_fw_dense240`	`te_residual_harmonic_gru_sequence_remote_Fw_dense240`	0.003219	Updated by this campaign
`residual_harmonic_gru_sequence_bw_dense240`	`te_residual_harmonic_gru_sequence_remote_Bw_dense240`	0.003492	Updated by this campaign
`residual_harmonic_gru_sequence_dense360`	`te_residual_harmonic_gru_sequence_remote_global_dense360`	0.003535	Updated by this campaign
`residual_harmonic_gru_sequence_fw_dense360`	`te_residual_harmonic_gru_sequence_remote_Fw_dense360`	0.003241	Updated by this campaign
`residual_harmonic_gru_sequence_bw_dense360`	`te_residual_harmonic_gru_sequence_remote_Bw_dense360`	0.003468	Updated by this campaign
`residual_harmonic_lstm_sequence_sparse_rcim`	`te_residual_harmonic_lstm_sequence_remote_global_sparse_rcim`	0.003368	Updated by this campaign
`residual_harmonic_lstm_sequence_fw_sparse_rcim`	`te_residual_harmonic_lstm_sequence_remote_Fw_sparse_rcim`	0.003234	Updated by this campaign
`residual_harmonic_lstm_sequence_bw_sparse_rcim`	`te_residual_harmonic_lstm_sequence_remote_Bw_sparse_rcim`	0.003440	Updated by this campaign
`residual_harmonic_lstm_sequence_dense240`	`te_residual_harmonic_lstm_sequence_remote_global_dense240`	0.003473	Updated by this campaign
`residual_harmonic_lstm_sequence_fw_dense240`	`te_residual_harmonic_lstm_sequence_remote_Fw_dense240`	0.003262	Updated by this campaign
`residual_harmonic_lstm_sequence_bw_dense240`	`te_residual_harmonic_lstm_sequence_remote_Bw_dense240`	0.003605	Updated by this campaign
`residual_harmonic_lstm_sequence_dense360`	`te_residual_harmonic_lstm_sequence_remote_global_dense360`	0.003477	Updated by this campaign
`residual_harmonic_lstm_sequence_fw_dense360`	`te_residual_harmonic_lstm_sequence_remote_Fw_dense360`	0.003351	Updated by this campaign
`residual_harmonic_lstm_sequence_bw_dense360`	`te_residual_harmonic_lstm_sequence_remote_Bw_dense360`	0.003556	Updated by this campaign

The family registries for the new Wave 2.3 model surfaces were created or refreshed by this campaign. The program-level training registry did not move to Wave 2.3 because the Wave 2.2 periodic `GRU` backward-only model still has the lower scalar test MAE.

TE Curve Verification Pipeline Boundary

`TE Curve Verification Pipeline` was not run as part of this closeout. Under the campaign governance rule, optional `TE Curve Verification Pipeline` verification remains a separate operator-approved workflow with a repository-owned launcher that can run locally or with `-Remote`.

Because Wave 2.3 does not beat the existing Wave 2.2 scalar best, a full `TE Curve Verification` refresh is not mandatory for accepting this closeout. If reviewed later, the forward-only sparse `RCIM` residual harmonic `GRU` is the only Wave 2.3 candidate that should be promoted into the optional verification queue first.

Closeout Decision

The campaign is complete and successful from an execution standpoint: all 18 runs completed, the leaderboard and best-run artifacts exist, and the family registries were refreshed.

From a modeling standpoint, Wave 2.3 is a completed comparison branch rather than a new best branch. The residual harmonic temporal structure remains available for future analysis, but the current project best should stay on the Wave 2.2 periodic recurrent family until official verification says otherwise.

Recommended Follow-Up

1. Keep the Wave 2.3 artifacts as a completed negative/neutral comparison branch.
2. Do not replace the current Wave 2.2 scalar best with the Wave 2.3 winner.
3. Run optional `TE Curve Verification Pipeline` only if visual curve behavior is worth inspecting despite the weaker scalar campaign result.